

Project Proposal - ECE 176

Abstract

We are implementing an exercise form corrector for 5 common bodyweight exercises (pushups, pullups, lunges, squats, and jumping jacks) using a deep learning algorithm on the UCF101 dataset. This proposal will cover the problem we are addressing, our key methods, and how we will test our solution's accuracy.

Problem Definition

Proper exercise form is critical for both safety and effectiveness. Performing exercises with bad form can place stress on certain joints or muscles, increasing the risk of injury. This can also lead to forming bad habits, which may be difficult to correct in the future. Many people lack access to a personal trainer or professional guidance due to financial or time constraints.

With the increasing popularity of at-home and bodyweight exercise, there is a need for personalized feedback to improve a user's exercise form. We want to offer a real-time solution that is accessible, simple to use, and easy to integrate into daily routine.

One of the main challenges in this problem is defining a standard for good form that the model can recognise. With the diverse characteristics of the human body, the model must be able to accurately handle different body types. Additionally, the model should also account for external factors such as lighting, camera angle, and clothing.

Overall, the problem we aim to solve is, "How can we design a real-time camera-based system that accurately analyzes people's bodyweight exercise form, and provide personalized and specific feedback to help them improve? "

Tentative Method

Overview

Our method will consist of three main components: pose estimation, form evaluation, and feedback generation. Video input from a camera will be processed to determine the exercise pose. Then its features will be analyzed to evaluate the user's form. Finally, feedback will be generated based on deviations from the expected exercise form.

Pose Estimation

We plan to use a feature extractor, such as Media Pipe Pose or MoveNet, to detect 33 keypoints on the body, including the shoulders, elbows, and knees, to classify poses. We will then train a deep learning classifier model, likely using a CNN model. This model will perform real-time classification, which makes it suitable for giving live exercise feedback.

Form Evaluation

Once we have extracted the key joints, we will compute features such as angles and distances relevant to each exercise category. For example, push-ups will need the elbow angle and back straightness to be measured. These features will then be compared against either learned or predefined standards with a certain amount of error to separate good and bad exercise form.

Feedback Generation

Based on the detected form errors, the system will generate real-time feedback to guide the user through printed messages. The feedback will be specific to the user's form, so that the user can implement it and improve their exercises.

Experiments

Dataset

UCF101- Action Recognition Dataset

- UCF101 contains 13320 videos of 101 different 'action categories'. For our purposes, we only plan to use 5 of the categories corresponding to the 5 bodyweight exercises mentioned above.
- We are using this dataset due to the diversity of examples it provides, which will help us determine an appropriate threshold for proper vs improper form.

Experiments

We plan to train a deep learning model (most likely a CNN) to classify good form versus bad form. We will test generalization by feeding it into an unseen subset from our dataset to ensure model robustness. Once our model has reached a strong classification accuracy, we will introduce classification by error-type. The model will determine whether the exercise being performed is done correctly, and if not, identify why and provide feedback to the user. Finally, we will implement our model live by testing it on ourselves/willing subjects and test its real-time accuracy by evaluating performance and user experience.

References

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